Lec-19

17/06/26 7:16 PM

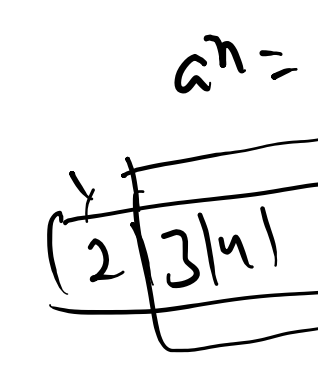
Nth Fibo

last Two term sum

N! = (n-1) — 1 Recursive call

(n) = (n-1) ''

$a^n = a^{n-1}$ ''



Recursion Tree

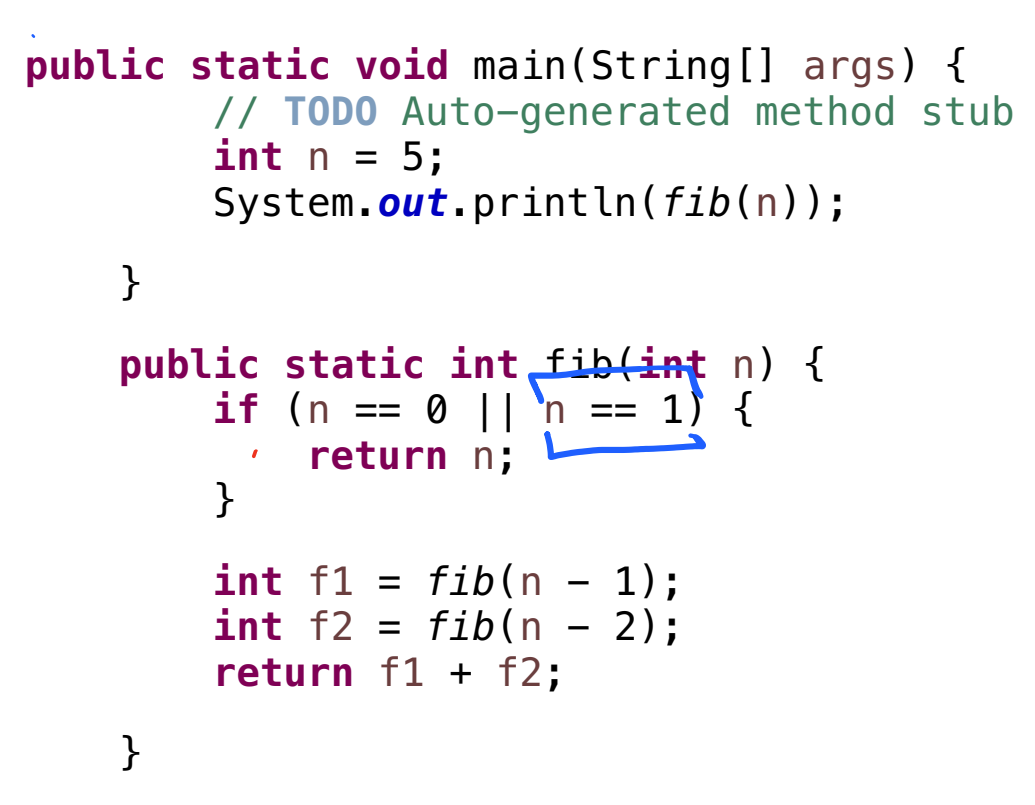
Nth

$f_n = f_{n-1} + f_{n-2}$

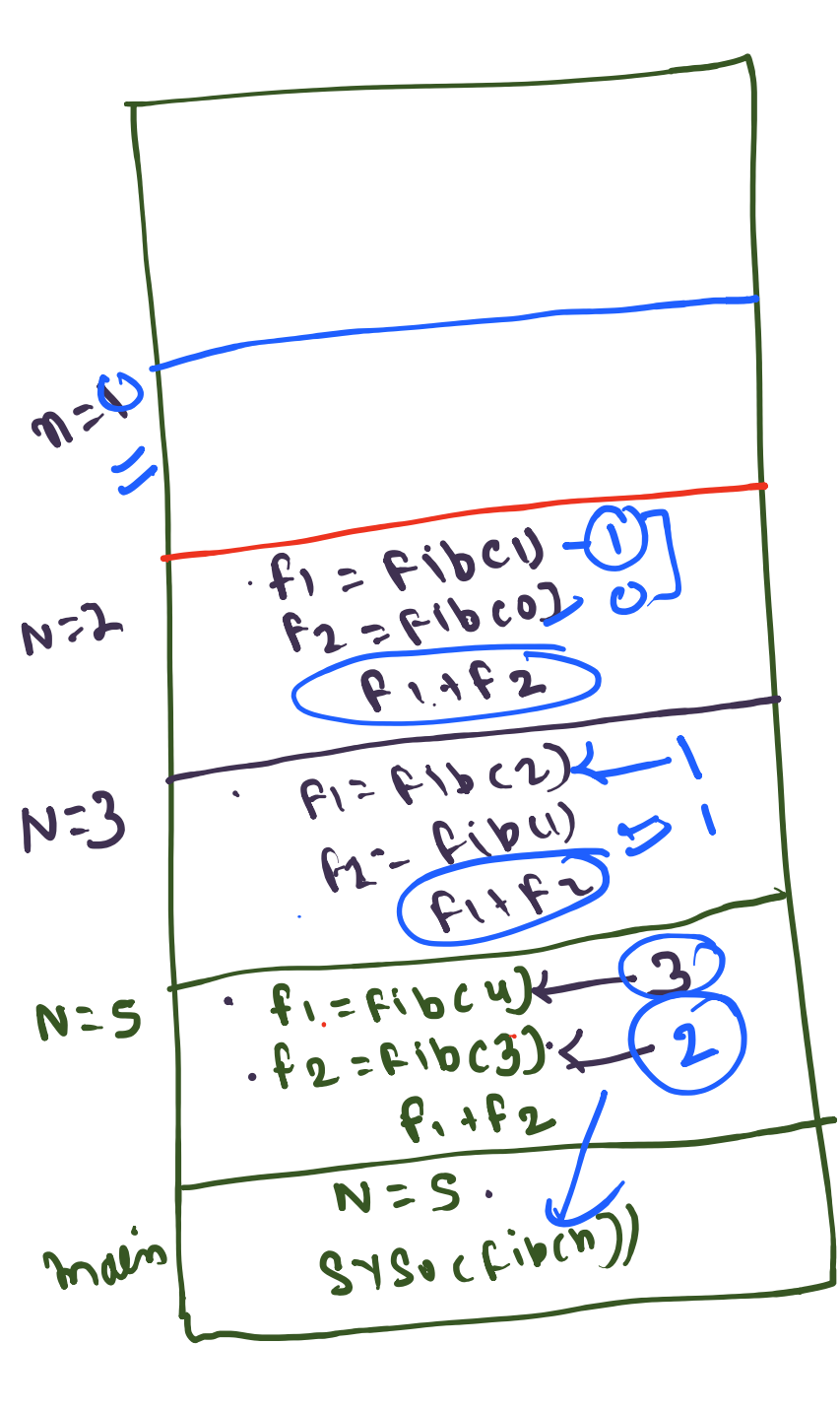
n=0 0 n=1 1

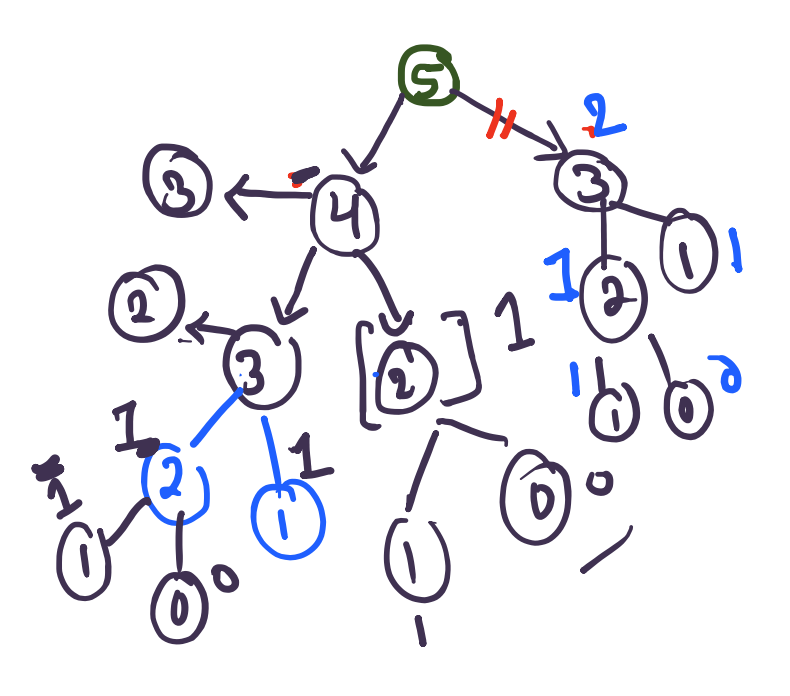
0 1 1 2 3 5 8 13 21 -

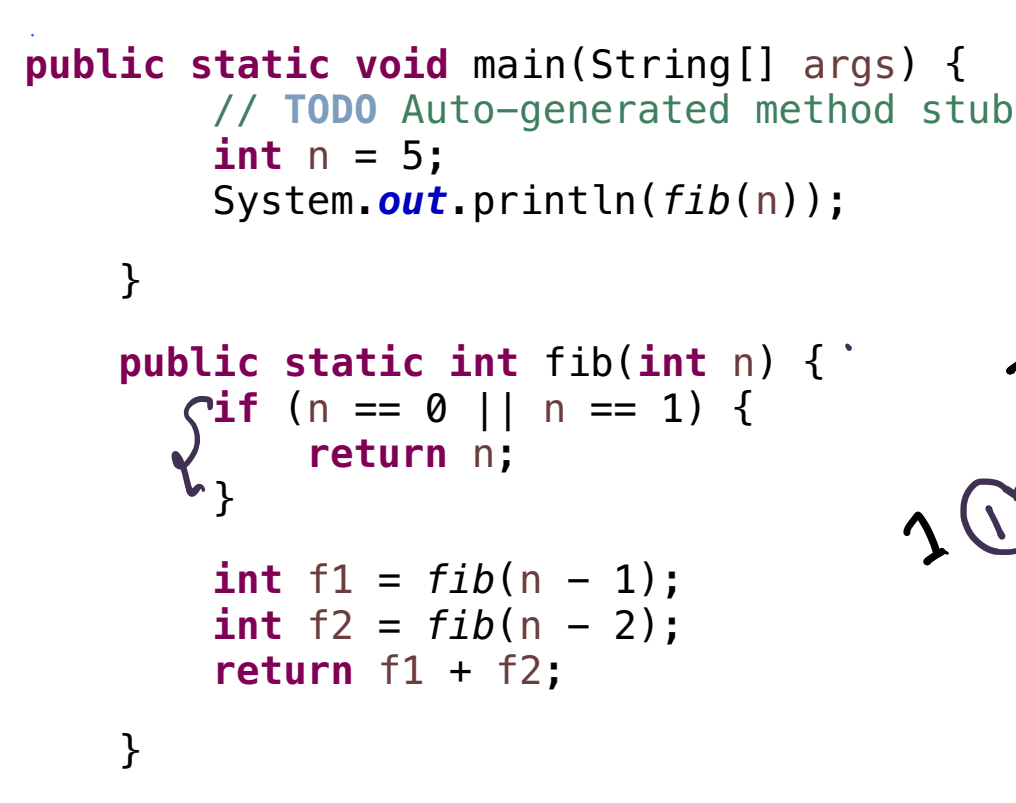
f1 = fib(n-1)
f2 = fib(n-2)
f1+f2



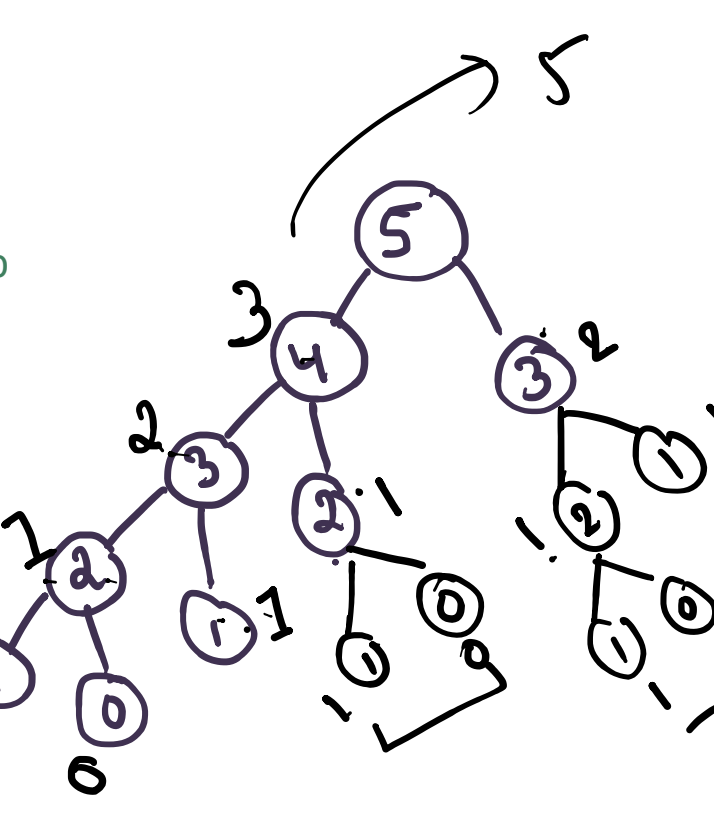
```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 5;
        System.out.println(fib(n));

    }

    public static int fib(int n) {
        if (n == 0 || n == 1) {
            return n;
        }

        int f1 = fib(n - 1);
        int f2 = fib(n - 2);
        return f1 + f2;

    }
```


n=0
N=2
f1 = fib(1)
f2 = fib(0)
f1+f2
N=3
f1 = fib(2)
f2 = fib(1)
f1+f2
N=5
f1 = fib(4)
f2 = fib(3)
f1+f2
main
N=5
sysout(fib(n))





```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 5;
        System.out.println(fib(n));

    }

    public static int fib(int n) {
        if (n == 0 || n == 1) {
            return n;
        }

        int f1 = fib(n - 1);
        int f2 = fib(n - 2);
        return f1 + f2;

    }
```


5
5
4
3
3
2
2
1
1
0

Subsequence

Sub Sequence

abc

Subsequence

Substring

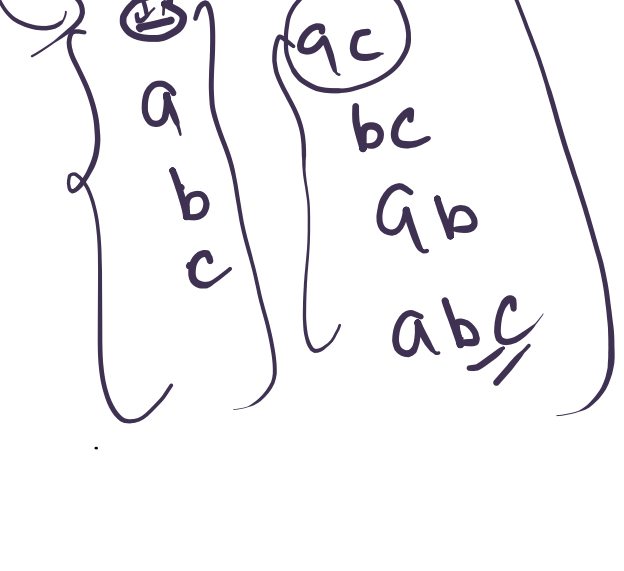
coding

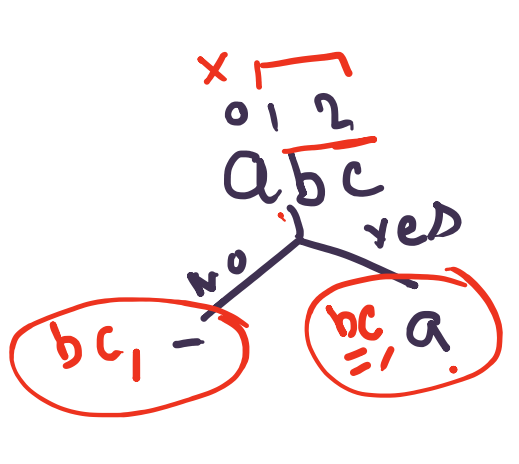
o ing ✓

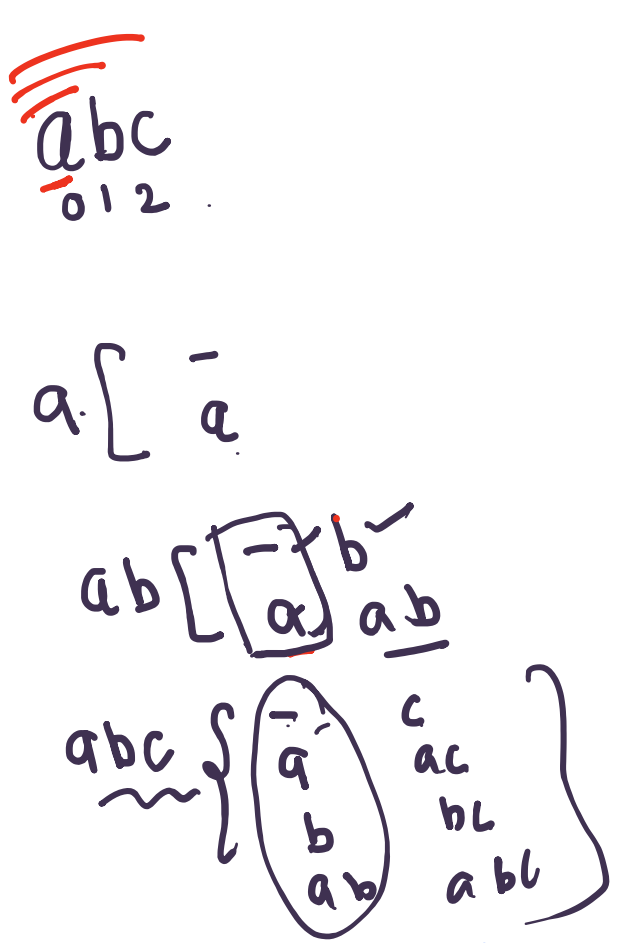
din ✓

Cdn

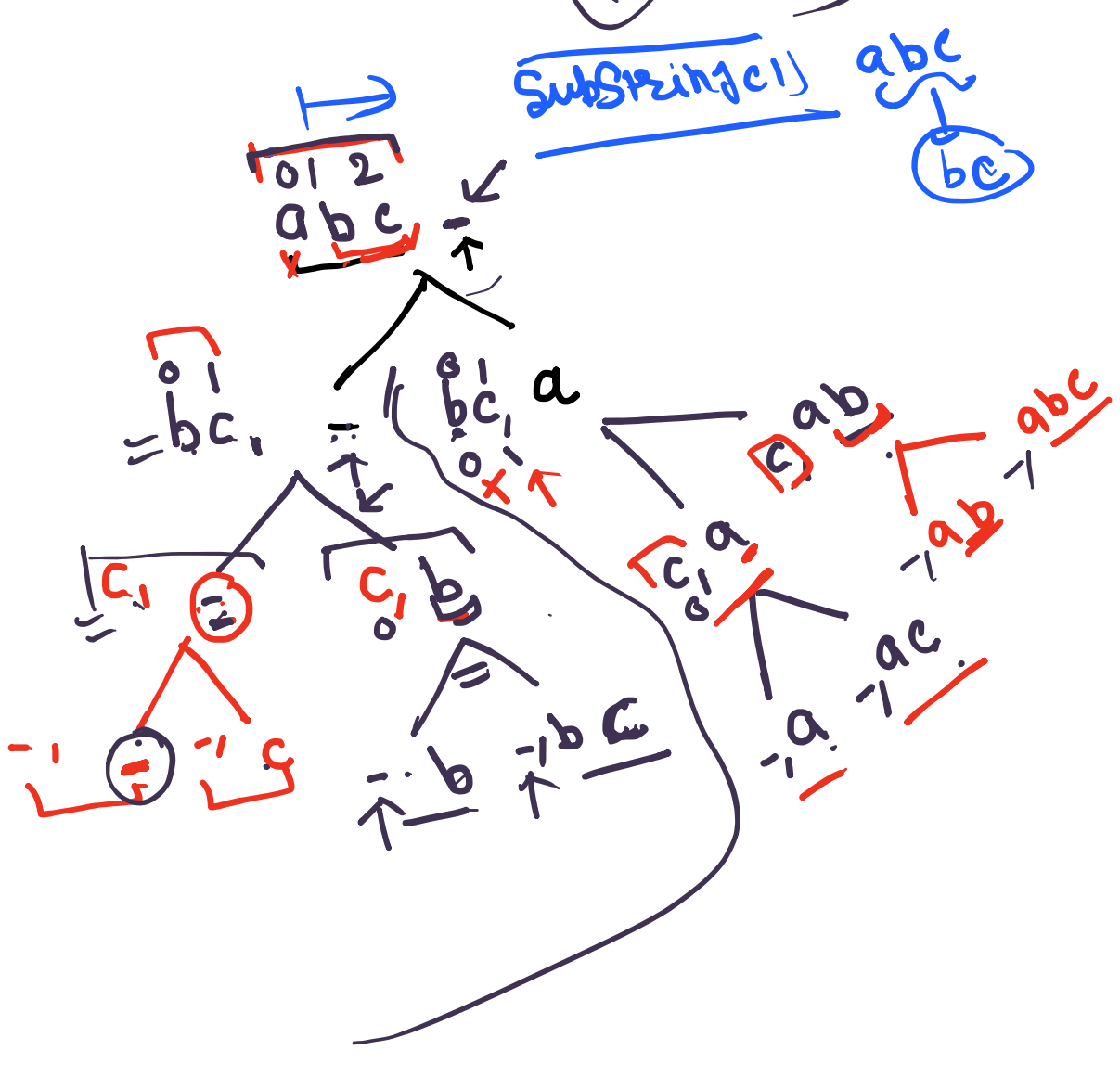
n -> [2^n]


abc
a
b
c
ac
bc
ab
abc

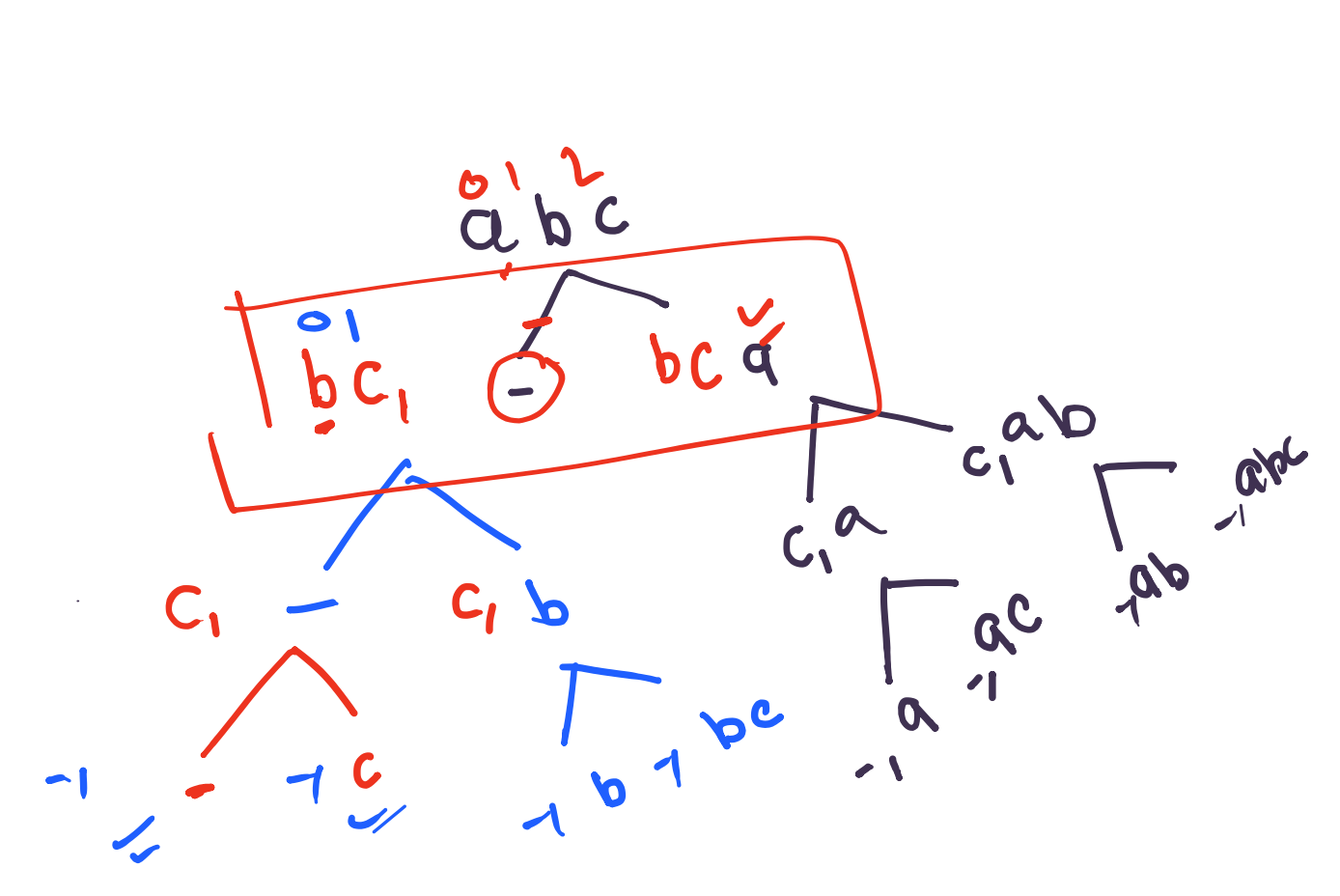

abc
012
no
yes
bc, -
bc, a

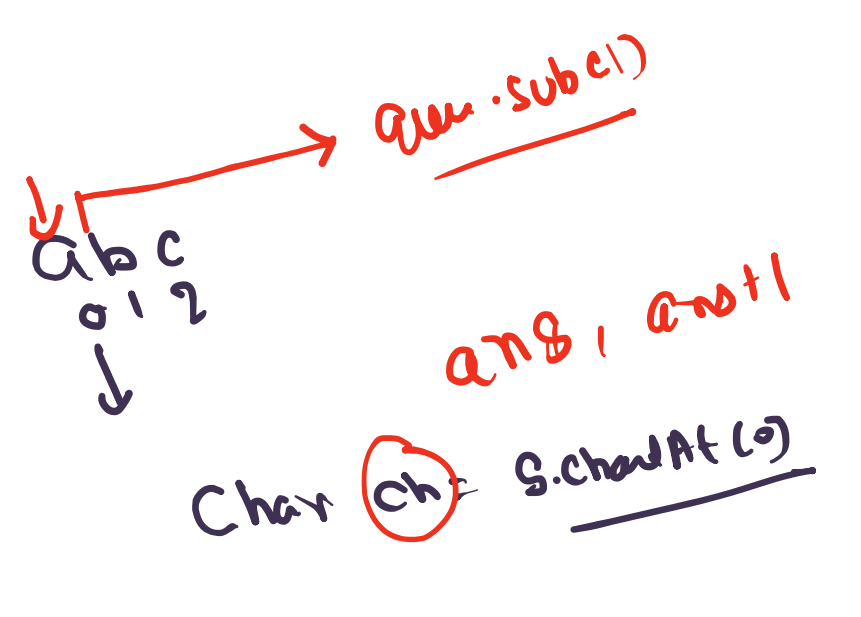

abc
012
a
a
ab
b
ab
abc
a
b
ab
c
ac
bc
abc

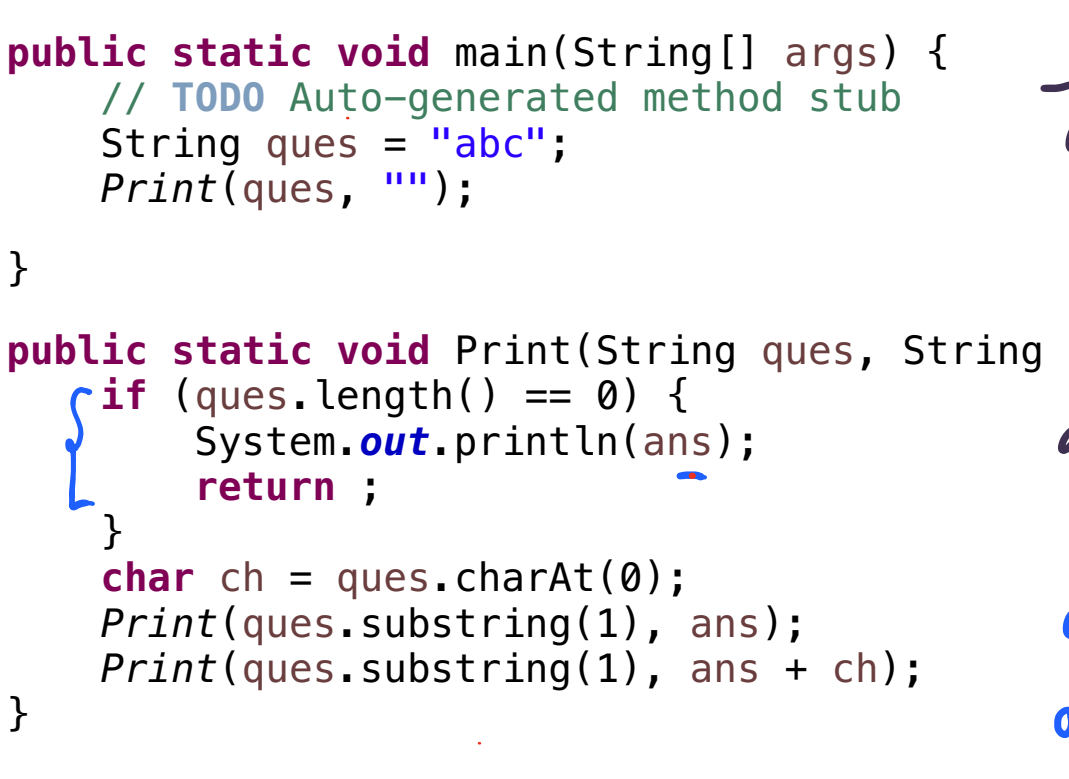
- c
b
bc
a
ac
ab
abc


Substring(1)
abc
bc
012
abc
bc, -
bc, a
c, -
c, b
c, a
c, ab
-, b
-, bc
-, a
-, ac
-, ab
-, abc

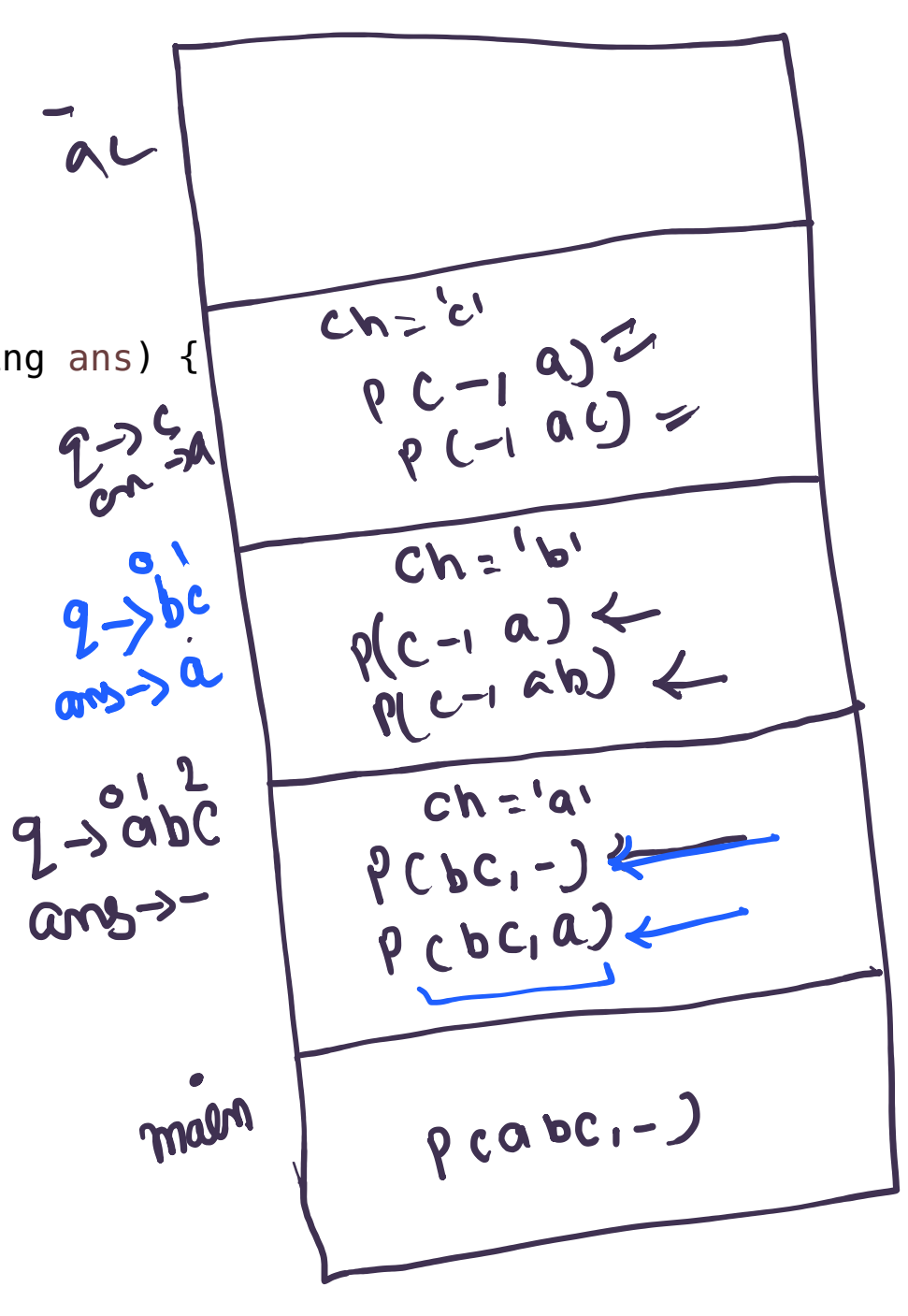
- c
b
bc
a
ac

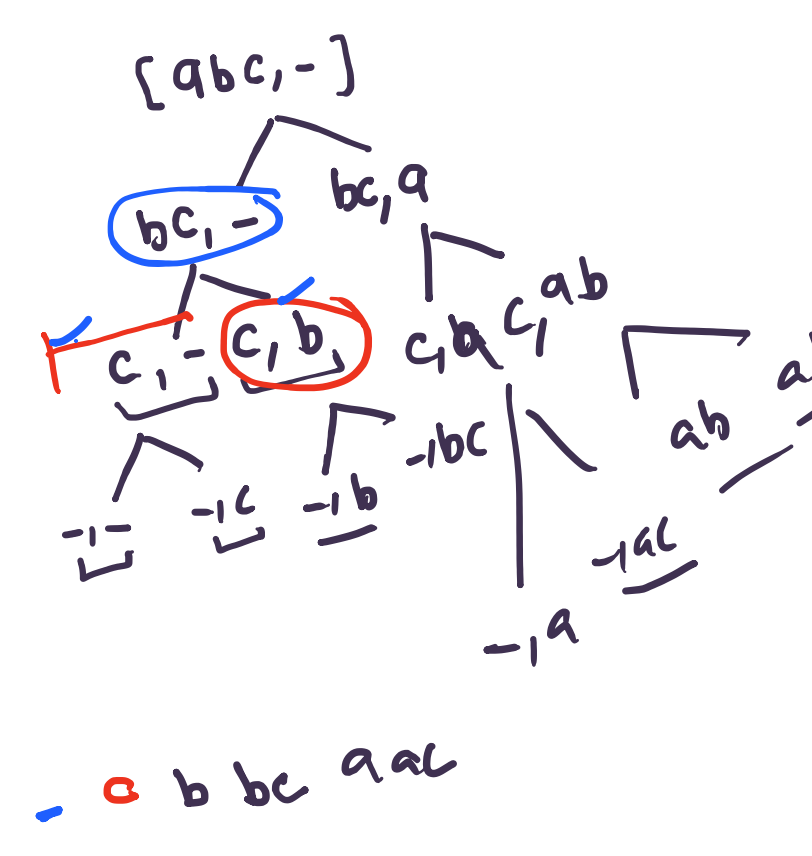

012
abc
bc, -
bc, a
c, -
c, b
c, a
c, ab
-, -
-, c
-, b
-, bc
-, a
-, ac
-, ab
-, abc


ques.sub(1)
abc
012
ans, ans+1
Char ch = S.charAt(0)

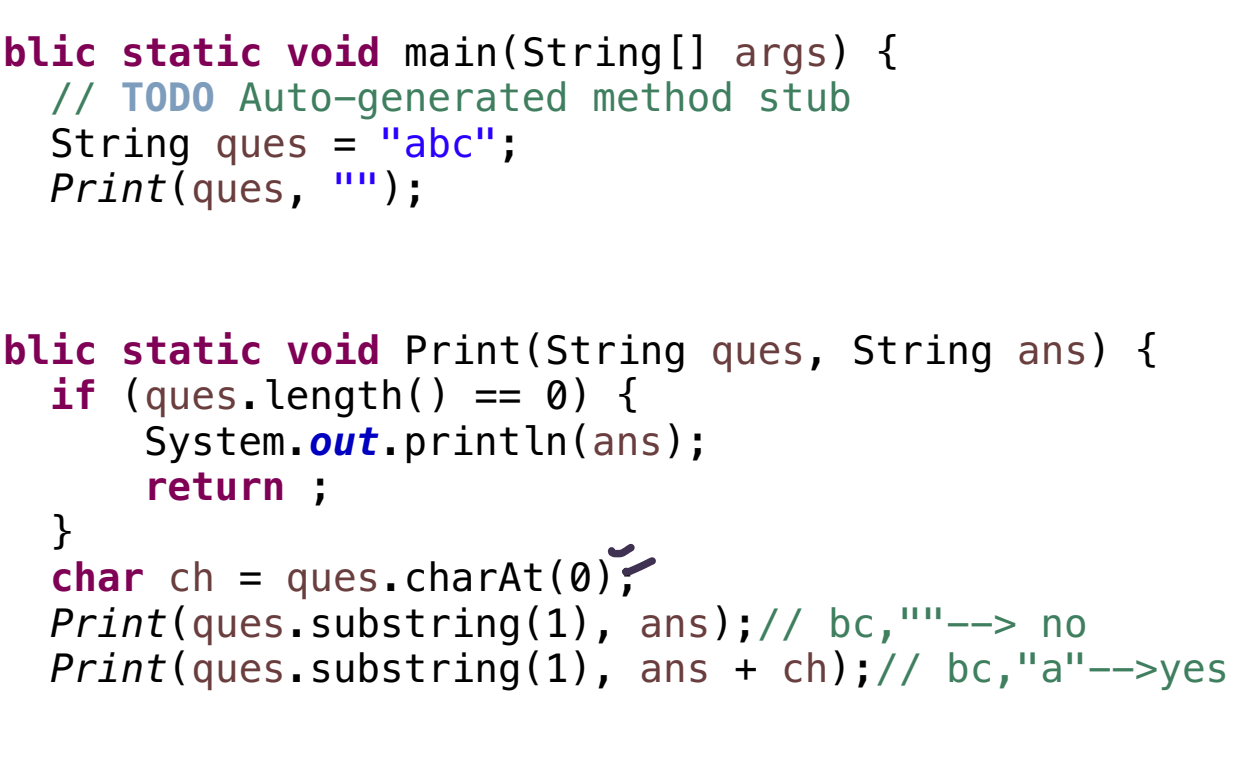


```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    String ques = "abc";
    Print(ques, "");

}

public static void Print(String ques, String ans) {
    if (ques.length() == 0) {
        System.out.println(ans);
        return ;
    }
    char ch = ques.charAt(0);
    Print(ques.substring(1), ans);
    Print(ques.substring(1), ans + ch);
}
```

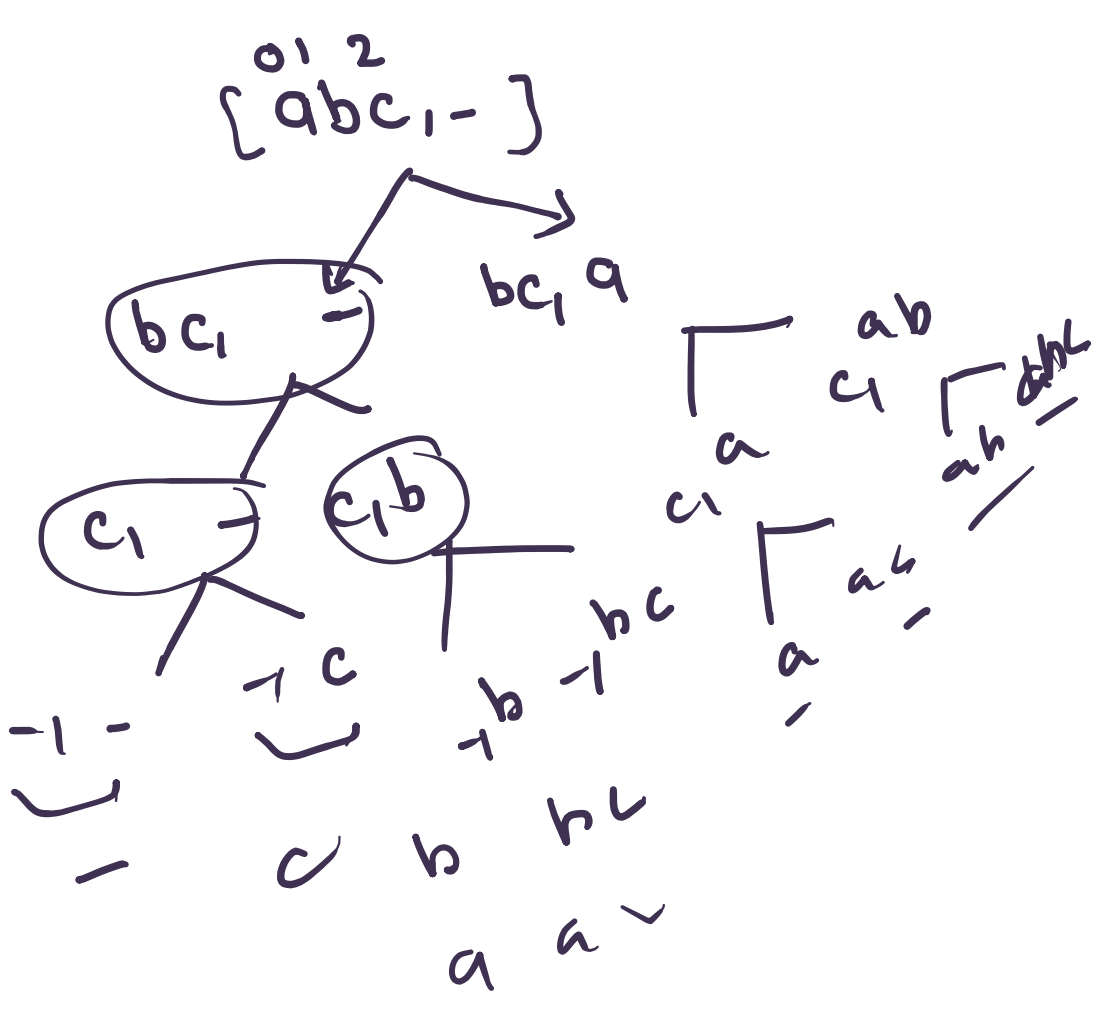

-ac
q->c
ch->A
ch='c'
p(c-, a)
p(-, ac)
q->bc
ans->a
ch='b'
p(c-, a)
p(c-, ab)
q->abc
ans->-
ch='a'
p(bc, -)
p(bc, a)
main
p(abc, -)


[abc, -]
bc, -
bc, a
c, -
c, b
c, a
c, ab
-, -
-, c
-, b
-, bc
-, a
-, ac
ab
abc

- a b bc a ac



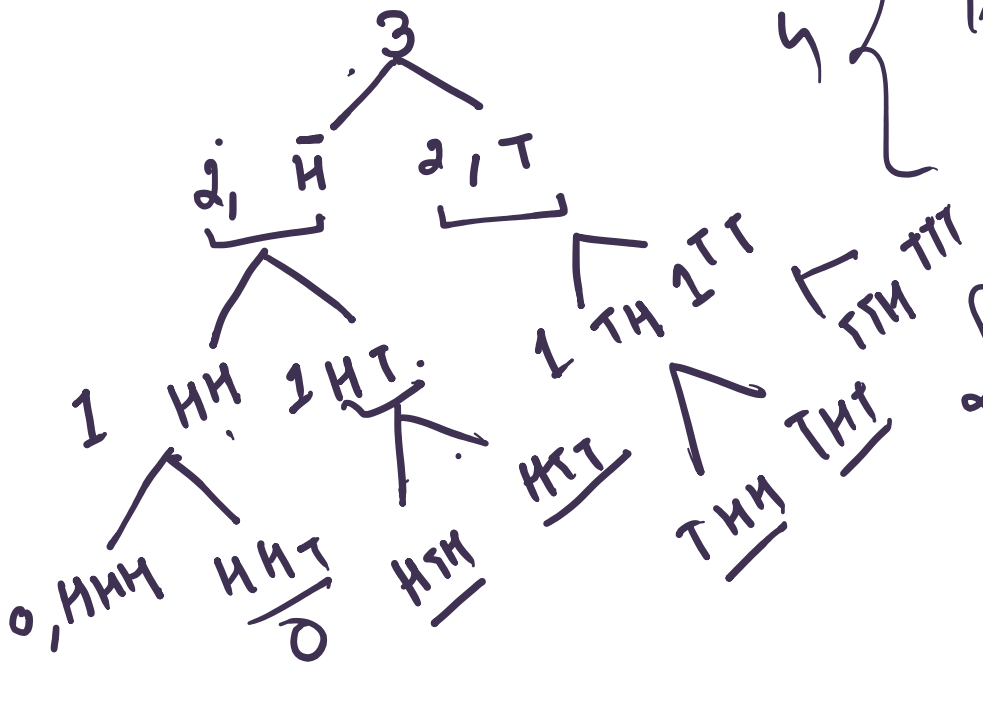
```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    String ques = "abc";
    Print(ques, "");

}

public static void Print(String ques, String ans) {
    if (ques.length() == 0) {
        System.out.println(ans);
        return ;
    }
    char ch = ques.charAt(0);
    Print(ques.substring(1), ans);// bc,""--> no
    Print(ques.substring(1), ans + ch);// bc,"a"-->yes

}
```


012
[abc, -]
bc, -
bc, a
c, -
c, b
c, a
c, ab
-, -
-, c
-, b
-, bc
a
ac
ab
abc
-
c
b
bc
a
ac

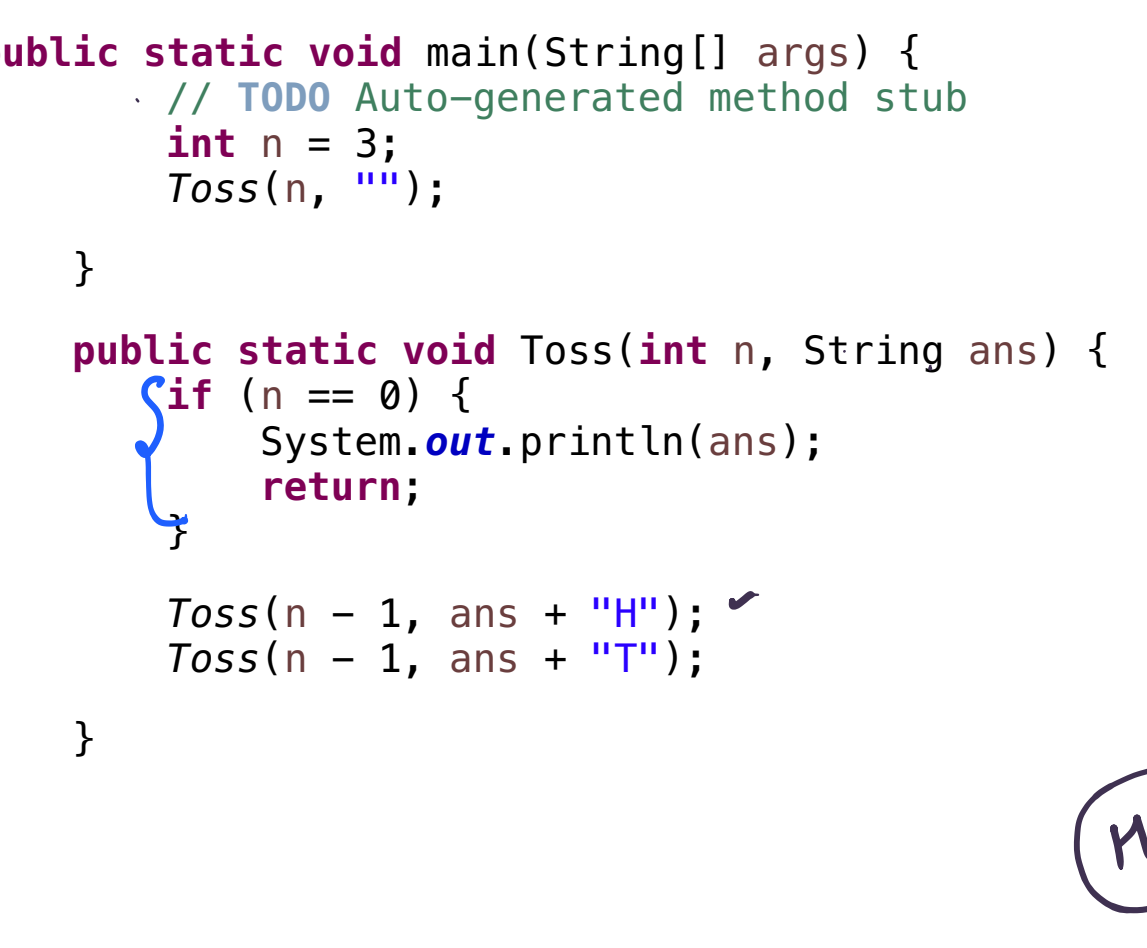
N-coins

JJP

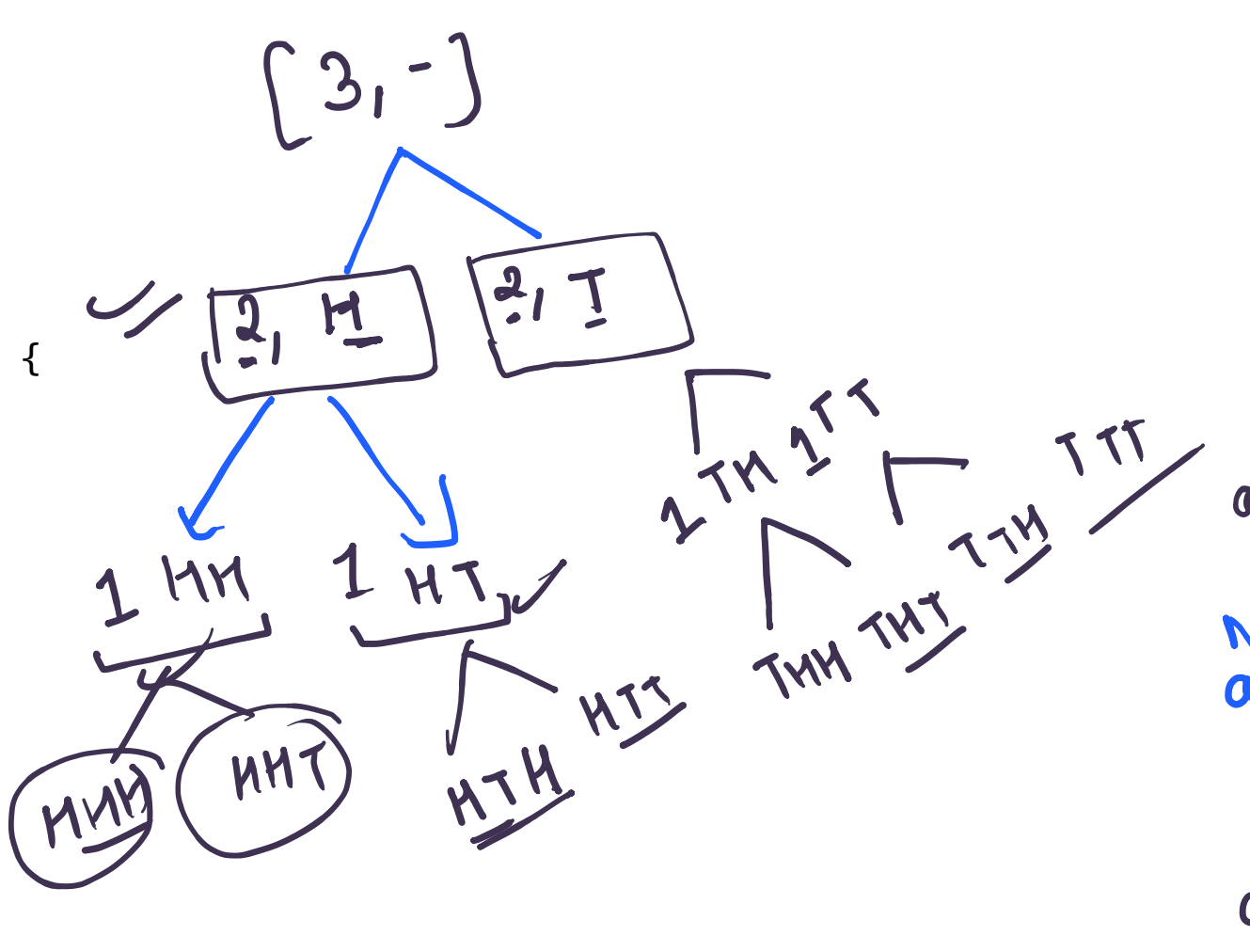
N=3

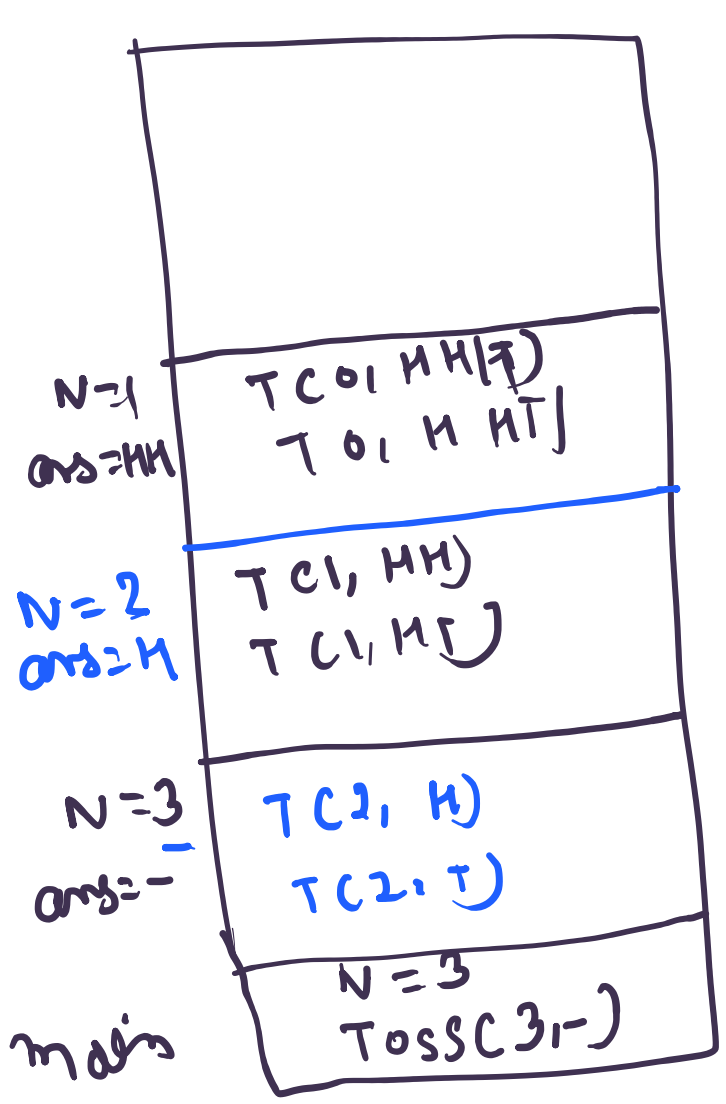

3
2, H
2, T
1 HH
1 HT
1 TH
1 TT
0, HHH
HHT
HTH
HTT
THH
THT
TTH
TTT
0

4 { HHH
HHT
HTH
HTT
THH
THT
TTH
TTT



```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 3;
        Toss(n, "");

    }

    public static void Toss(int n, String ans) {
        if (n == 0) {
            System.out.println(ans);
            return;
        }

        Toss(n - 1, ans + "H");
        Toss(n - 1, ans + "T");

    }
```


[3, -]
2, H
2, T
1 HH
1 HT
1 TH
1 TT
HHH
HHT
HTH
HTT
THH
THT
TTH
TTT


N=1
ans=HH
T(0, HH(T))
T(0, H HT)
N=2
ans=H
T(1, HH)
T(1, HT)
N=3
ans=-
T(2, H)
T(2, T)
main
N=3
Toss(3, -)